Lab: Masks

In past few labs we've been animating still images using Adobe After Effects, in this lab we will cover working with videos in a motion graphic project. This includes editing videos, applying masks and filters.

In this lab we will build a video advertisment for the fictious island of St. Zazou. The advertisement will show beautiful beaches and palm trees to show off the natural beauty of the island, we have a concept of a fun, creative video wall effect.

Setup

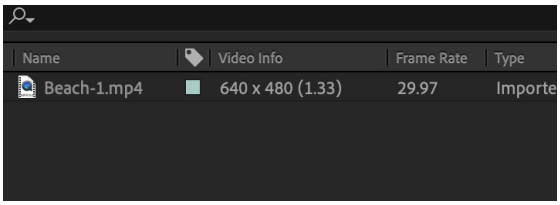
Take a moment to look at the video will will create using Adobe After Effects. You can view the video on YouTube below:

https://youtu.be/Hamh2IoJ3TI

Make sure you have downloaded the lab files needed to complete this lab. The files are in zip folder called *After-Effects-Video-Masks.zip*. Make sure to unzip the folder completely to use the files inside appropiately.

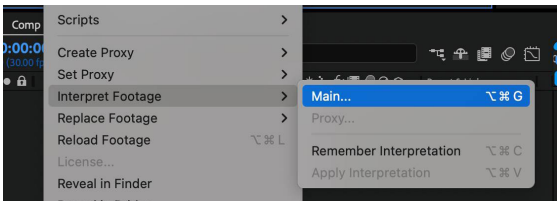
Videos

In addition to working with images, After Effects can also work with videos. After Effects can import most standard video formats including Quicktimes movies *(.mov), MPEG4,* and Microsoft AVI formats.

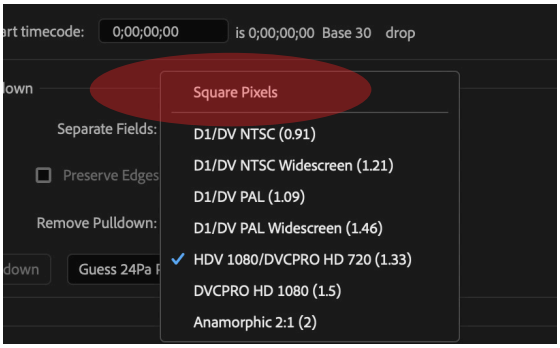


Create a new project in Adobe After Effects and import the *Beach-1.mp4* video from the lab folder into your *Project* panel.

It is helpful before working with videos in a motion graphic project to know a bit about the videos properties such as the dimensions and frame rate. In the *Project* panel use the *Video Info* column to view the dimensions, aspect ratio and the frame rate of the *Beach-1* video. You should see that the video is *640 x 480px* with a framerate of *29.97*.



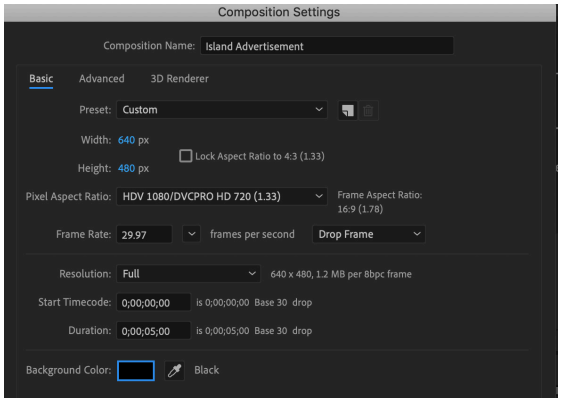
If you view the *Beach-1.mp4* video you may noticed it says *1.33* next to the dimensions, that is the pixel aspect ratio. What this means is that when the camera recorded the lense captured more pixels in the horizontal direction, than the vertical. The issue with this is most digital programs use square pixels or a ratio of 1 not 1.33.



We can fix this issue by selecting the file in the project folder and going to *File > Interept Footage > Main*. This will bring up a dialogue box that allows you to change the settings for the video.

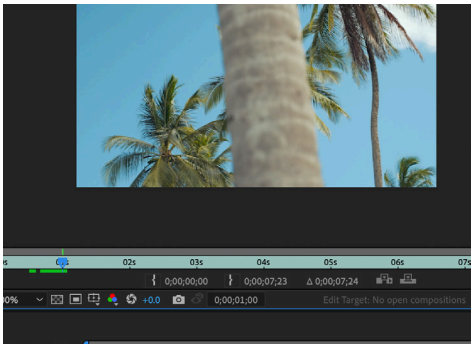
Go to *Other Options* and set *Pixel Aspect Ratio* to *Square Pixels*.

Import *Beach-2.mp4*, *Beach-3.mp4* and *Beach-4.mp4* into the project. Repeat the *Interept Footage* option for each and set it *Square Pixels*.



Edit the Video

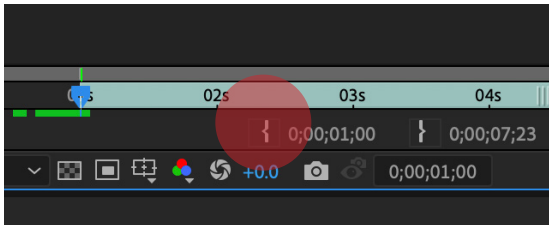
Let's create a composition called *Island Advertisement* by going to *Composition > New Composition*. Set the dimensions of the composition to 640 x 480px, give it a *framerate* of *29.97* and duration of 5 seconds or 00:00:05:00.



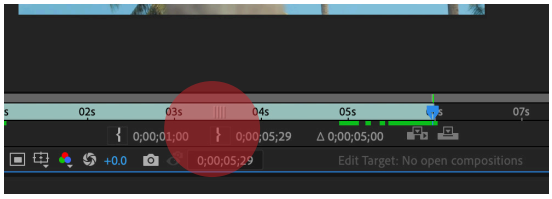
Double click on *Beach-1.mp4* in the *Project* panel to open the video into the *Footage* tab of the *Composition* window.

The *Footage* window allows you to preview your video before bringing it into your composition timeline. Use the *spacebar* to play the video and you should see a black background with title text followed by a moving shot of a palm tree and then more black background with title text.

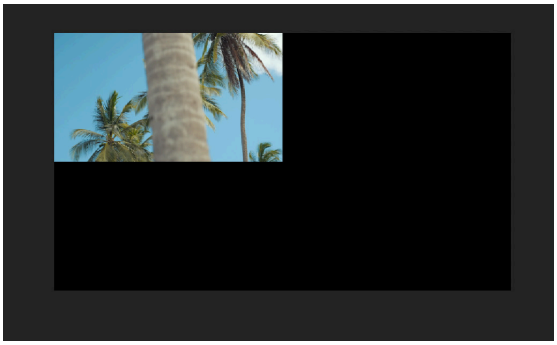
We don't necessary want the entire piece of footage for the final composition, so let's edit the video and trim it down to just the part that we want. To edit the video we will use *in* and *out* points. *In* and *out* points set the *"in"* or beginning of the part of footage we want and the *"out"* or the end of footage we want to use.



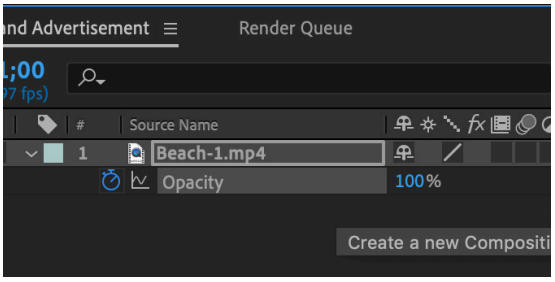
Use the *timeline* in the *Footage* window and set the time indicator to 1 second in, or *00:00:01:00*. This is the beginning of the footage we want to use, so let's set an *in point* here. Use the *Set In* icon at the bottom of the *Footage* window to the set an in point. You'll notice that the light green footage line gets shorter and now start at the 1 second mark. Move the time indicator to 6 seconds or *00:00:06:00*, use the *Set Out* icon to place the *out point*. Once again you'll see the light green footage line get shorter. This is now your edited footage.



If you like you can always reset your *in* and *out* points. Simply move the time indicator along the timeline and hit the *Set In* and *Set Out* buttons again to create new in and out points for the video.



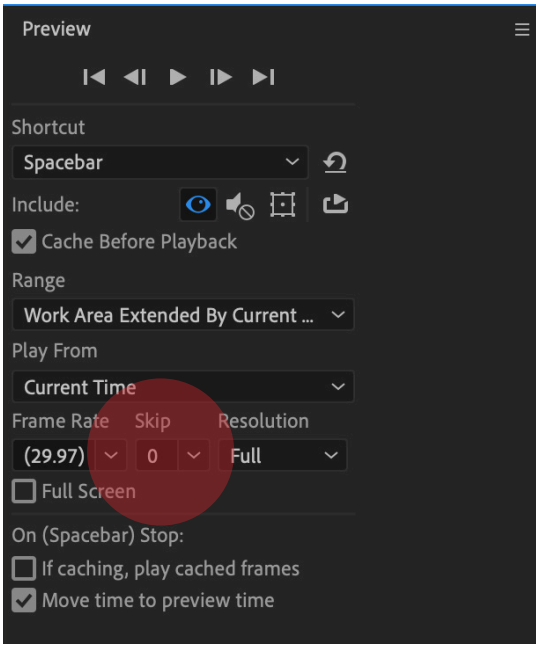
Click back on the *Composition* tab in the *Composition* window to view the composition timeline. Click and drag *Beach-1.mp4* into the *Composition* timeline. Play the composition to see that the footage now begins with the moving shot of the palm tree.



Animate Videos

Now that we have an edited video in our composition let's play around with it. You'll notice that the *Beach-1.mp4* video takes up

the entire composition. What we want to create a wall of multiple videos, so to accomplish this let's scale down the size of *Beach-1.mp4*. Use the *scale* property for the *Beach-1* layer and scale it *50%* for height and width. Use the *Selection Tool* to place the video at the top left corner on the composition screen. Your composition should look something like the one on the right.



We can animate properties of a video just like we animated properties of an image in previous labs. Let's animate the *opacity* property so the beach video fades in to our scene.

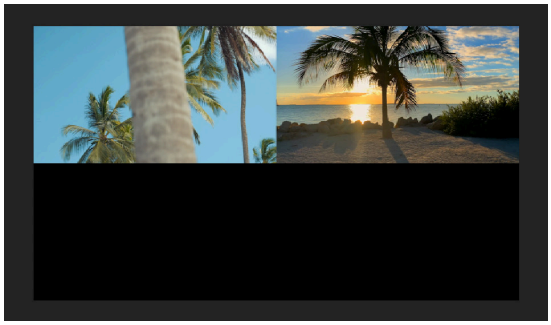
Use the *opacity* property and keyframe it so the opacity goes from 0% to 100% from 00:00:00:00 to 00:00:01:00. Save and preview your timeline to see the results; the results will be that the *Beach-1* video fades into the scene.

You may notice that previewing an animated video takes much longer for the computer to render than previewing an animated image. This is because videos are much more intensive on the processor of the computer, so it takes a bit more time to render the whole composition in real time.

One way to speed up the process of rendering a preview is to not render every frame, but every other frame. Previewing every other frame will be less work for the computer and therefore the video will render and preview faster.

To render every other frame for preview open the *Preview* panel in Adobe After Effects. If it is not already open in your setup go to *Window > Preview*. In the Preview panel there is an option called *Skip*. *Skip* means "Should the preview renderer skip any frames while previewing?" At the moment it is set to *0* meaning the preview renderer does not skip any frames, or renders all of the frames. Let's set it to *1*, which means for every frame the computer renders it will skip 1 frame; or in other words it will only render every other frame. Now when you renderer every other frame the process of previewing your composition will go faster. The one downside of this option is the video in the preview will be a little bit choppy as you are only seeing every other frame. Don't worry, as this only appears in the preview of the program, but not your will video export. In the final video export After Effects will render every frame.

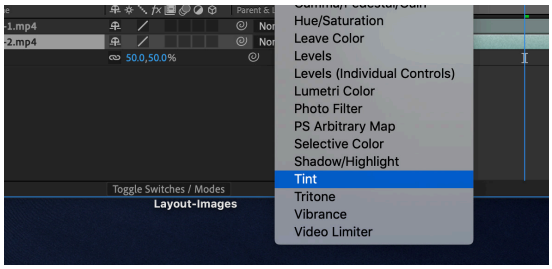
Add Another Video & Apply Filters



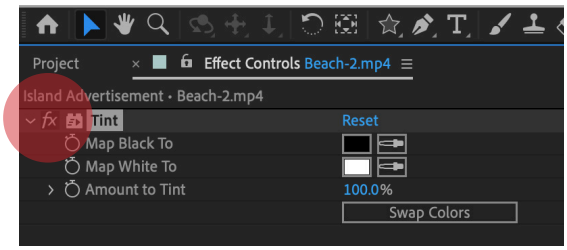
Import *Beach-2.mp4* into the project. Use the *Footage* window to edit out the black ground and title text. In the end you should only have footage of the palm tree looking over the ocean.

Add *Beach-2.mp4* to your composition. Use the *scale* property

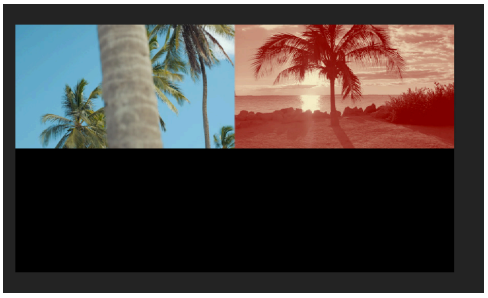
to scale the video to *50%* and place it in the upper right hand corner. Use the image on the right as a reference.



Let's add a little bit of pizazz to this video. Let's add a *filter* to change the look of the video, in this case we'll use the tint option. Select *Beach-2.mp4* layer and go to *Effect > Color Correction > Tint* to apply a *tint* effect to video. When you apply the filter you'll notice a new tab open up to the right of the *Project* panel called the *Effects* tabs. The *Effects* tab shows all of the filters applied to a layer. You should see the tint effect being applied.



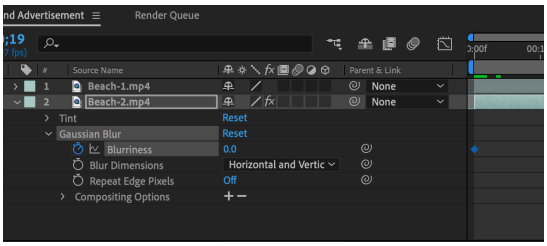
The filter can be turned on and off by using the *FX* checkbox in the upper left hand corner. Different filters apply different effects and how each effect work depends on the effect itself. In this case we have are using the *tint* effect. The tint effect works by setting your the tones of the video to two different colors. The *Map Black To* sets the color for the dark tones in the video and the *Map White To* sets the coor for the light tones in the video. Let's set *Map Black To* to a darker red, like a crimson, and set *Map White To* a light yellow color. The result should be something like the image on the right, don't worry if yours is not 100% the same.



Animate Effects

In addition to adding filters to layers, we can also animate them as well. We are going to add a second filter to *Beach-2.mp4* video. We are going to animate the video starting off blurry and then becoming clear.

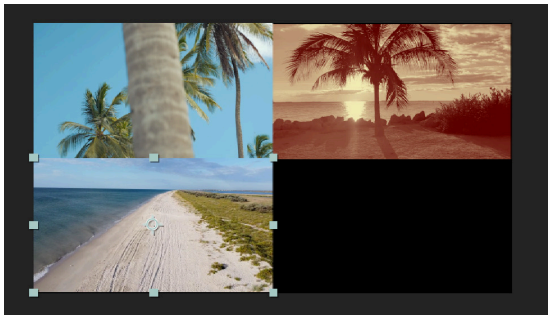
Select the *Beach-2.mp4* layer and go to *Effect > Blur & Sharpen > Gaussian Blur* to apply a Gaussian blur to the layer. Take the time indicator and drag it to the beginning of the timeline. In the *Effects* tab set the *Gaussian Blurriness* option to *50*. Next to the *Blurriness* option in the *Effects* you will see a stop watch icon just like the one we used to animate properties like position, rotation and scale.



Click the stop watch and you will see a circle appear in the *Composition* timeline, this is a keyframe for the *blurriness* property. To see the filters and keyframes applied to the video in the timeline hit the *"E"* key on your keyboard, *"E"* for effects. You will see it lists both the *tint* and *Gaussian* filters in the timeline. Unfurl the *Gaussian blur* layer and you will the *Blurriness* property with a keyframe applied to it. Move the composition time indicator to the 1 second mark or *00:00:00:01*. Change the *Blurriness* property from *50* to *0*. Preview and play your video and you should see the beach video start blurry and then become clear.

Save your Project

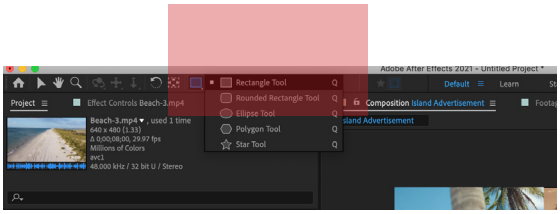
Add Layer Masks



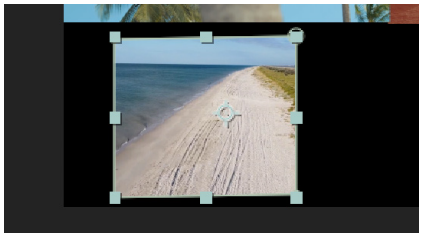
Let's add another video to our video wall we are creating for the project. Import *Beach-3.mp4* into the project and add it to our composition. Scale the video to *50%* for height and width and place it in the lower left hand corner. View the image on the right as a reference.

If you see the original video you can see that the video in the lower left is a square shape. Our original video was in a 4:3 or rectangle aspect ratio, so the video will always be a rectangle, even if you scale it. So how do we solve this issue? The answer is *layer masks*.

Layer masks allow you to hide part of your footage. Think of it like a Halloween or mardi gras mask; when you wear the mask your face is concealed from others, but at any time you can simply take the mask off to reveal your face. We will do the same thing but with a layer mask.

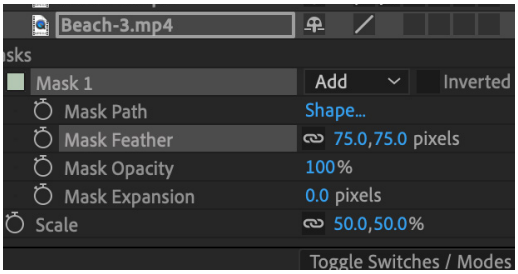


Begin by selecting the *Beach-3.mp4* layer, In the *Toolbar* select the *Rectangle Tool*. Drag the *Rectangle Tool* over the video in the *Composition* window, you should see an icon appear that looks like a rectangle with a hole in it. As you drag you should see part of your video disappear, congratulations, you've created a mask.



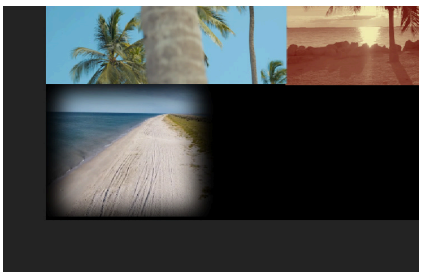
To access the mask you created click the *Beach-3.mp4* layer and press the letter *"M"* on the keyboard, *"M"* for *masks*. You should see a layer labelled *Mask1*, unfurl this to the see the different options for the mask. To remove a mask you added to a layer select the *Mask* in the timline and hit *Delete / Backspace* key on your keyboard.

To change the size of the mask select the *Beach-3.mp4* layer and double click on the mask using the *Selection Tool*, this will open up a free-transform box that allows you to resize your mask. Hit *Return* or *Enter* to confirm the mask resizing.



In additional to resizing the mask you can also pan the video inside of the mask. Select the *Pan Behind* tool click and drag on the video and notice you can view different areas of the video.

At the moment the mask or cut away from the video is very crisp, let's soften the edges of the mask to give it a more whispy quality. Select *Beach-4.mp4* and press *M* to get the mask properies. Unfurl *Mask 1* and find the *Mask Feather* option. The *Mask Feather* option is how feathered or blurred the edges of the mask. Defaultly masks are set to a *feather* of *0*, or crisp edges. To increase the blurred edge set the *Mask Feather* to *35px*.

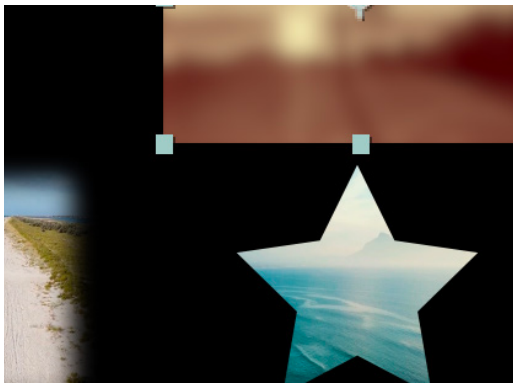


Render and preview the composition. You should see the video with soft edges as the compositon plays.

Save your project

Make an Additional Mask

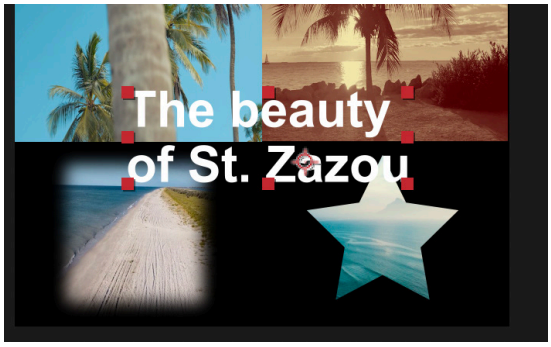
Import *Beach-4.mp4* into your project and add it to your composition timeline. Use the *scale* property to scale the video to *50%*. Use the *Selection Tool* to place the video in the lower right hand corner of the composition window, use the image on the right as a reference.



In the last section we applied a basic mask to our video. In addition to rectangles and squares you can create other masks as well. With the *beach-4.mp4* layer selected click and hold on the *Rectangle Tool* to get a dropdown menu. You will see options such as ellipse, polygon and star. Select the *Star Tool* and click and drag above the video to create a star mask, you video should look something like the image on the right.

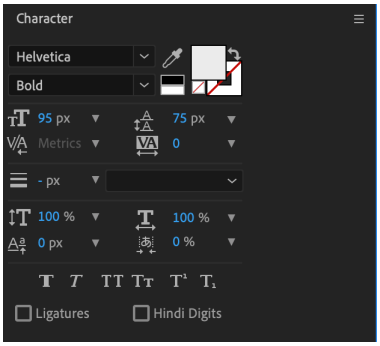
Add Type

Now that we have our video wall with different videos let's add tet for our advertisement. Select the *Type Tool* from the *Toolbar* and click in the center of the composition screen. Type the following phrase:




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The beauty
of St. Zazou
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Now that we have our text let's format is using the *Character* and Paragrpah panel. Use the *Type Tool* to highlight the text. Open the *Character* panel or go to *Window > Character*. The *Character* panel allows us to modify text properties such as typeface, font size, leading and other properties.

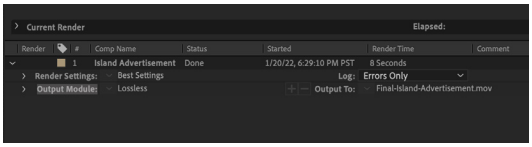


Set the font size of the text to *95px* and set the *font-family* to *Arial bold*. Let's also adjust the space between our lines of text, this is referred to as leading. Set the *leading* to *75px* to make the lines tighter with one another.

Finally, let's set the alignment of the type to center the block of type. To do this we'll use the *Paragraph* panel. Open the *Paragraph* panel or go to *Window > Paragraph*. In the *Paragraph* panel in the alignment option choose the center text option. Render and preview your animation

Export





Great job! Now let's export the final project to show the client. With the *Island Advertisement* composition open in the timeline go to *Composition > Add to Render Queue*.

In the *Render Queue* look for *Comp Name* make sure it is set to *Island Advertisement*.

Under *Render Settings* make sure it is set to *"Best Settings".*

Under *Output Module* make sure it is set to *"Lossless"* make sure the format is set to either Quicktime or AVI.

Finall, use the *Output To* option save the animation to your Desktop with the label *lastname-Animation-filter-mask.*

Submission

Upload your exported project video to your YouTube account. Publish the video and you will copy and paste the video URL to the submission page for the lab.

In addition to the final video url please upload the *lastname-advanced-filter-mask.aep* file to Canvas for grading. Please do NOT submit all of the files or videos as those are large files, instead just upload the *.aep* project file for grading purposes.

Extra Credit: Not Required

+2 points

The following is extra credit and is not required as part of the lab.

Spice up the advertisement by animating the text.

- Apply a Gaussian blur filter to the text. Have the text start very blur at the beginning of the project and become clear or no blur after 1.5 seconds.
- Keyframe and animate the opacity property of the text layer to have the text fade up from invisible to fully visible after 1.5 seconds.